

To determine the amount of RAM necessary for the virtual machine (VM) portion of this project, you should consider the specific roles of the VMs you are creating and the overall specifications of your host machine.

Recommended RAM for Lab VMs

The sources suggest the following memory allocations for the different components of the SOC lab:

- **Ubuntu Server (Endpoint):** A standard recommendation is **4GB of RAM** 1. however, depending on your host machine's specifications, you can drop this as low as **2GB** if necessary 1.
- **Windows 10 VM (Endpoint):** It is recommended to allocate **4GB of RAM** to ensure the Windows environment runs smoothly during the setup and deployment of agents 2.
- **Wazuh Manager:** While the specific snippets provided focus on the endpoints, the process uses the same installer image for the Ubuntu server that acts as the manager, implying a similar baseline of at least **4GB** is preferred for stable performance 1.

Factors for Determination

- **Host Machine Specs:** Your total available physical RAM determines how much you can distribute among your VMs 1. If you have limited resources, prioritize the **2GB minimum** for Linux nodes 1.
- **Dashboard Verification:** Once your agents are connected and active, you can monitor their actual resource usage directly. By navigating to "**View agent details**" in the Wazuh dashboard, you can see real-time data on how many **cores and how much memory** the agent is actually utilizing 3.
- **Performance Requirements:** Using **Remote Desktop Protocol (RDP)** to manage your Windows VM is noted as being "much more easier to work with" than the standard VMware Workstation console, but it may require the VM to have enough RAM (the recommended 4GB) to handle the connection and background services like Sysmon effectively 2, 4.

In summary, you should aim for **4GB per VM** for an optimal experience, but you can scale down to **2GB for Linux-based machines** if your hardware resources are constrained 1, 2.